

TrustRAG: An Adaptive, Explainable Multi-Agent Retrieval Framework for Intelligent Document Analysis

ABSTRACT:

The rapid growth of digital documents across academic, industrial, and enterprise environments has made efficient knowledge retrieval and document understanding increasingly challenging. Although Retrieval-Augmented Generation (RAG) has significantly improved document-based question answering, existing RAG systems still suffer from static retrieval strategies, hallucinated responses, limited explainability, and inadequate verification of generated content. To address these limitations, this work proposes TrustRAG: An Adaptive, Explainable Multi-Agent Retrieval-Augmented Framework for Intelligent Document Analysis, an intelligent platform that enables users to upload heterogeneous documents such as PDF, DOCX, TXT, technical manuals, reports, and research papers, and interact with them using natural language queries. The proposed framework introduces an Adaptive Retrieval Selection Module that dynamically selects the most suitable retrieval strategy—BM25, Dense Retrieval, or Hybrid Retrieval—based on the user's query to improve retrieval relevance and contextual accuracy. A LangGraph-based Multi-Agent Architecture coordinates specialized agents for document understanding, question answering, summarization, comparison, citation generation, evidence verification, and report generation. An Evidence Verification Module validates generated responses against retrieved evidence to reduce hallucinations, while a Confidence-Aware Explainability Module provides supporting passages, confidence scores, and source citations to improve transparency and user trust. The framework is implemented using React, FastAPI, Gemini/OpenAI LLMs, BAAI BGE Embeddings, Qdrant, and PostgreSQL within a scalable microservice architecture. The proposed approach will be evaluated against conventional RAG systems using Precision@K, Recall@K, MRR, nDCG, Faithfulness (RAGAS), Answer Relevancy, Hallucination Rate, and Response Time. By integrating adaptive retrieval, evidence verification, confidence-aware explainability, and multi-agent orchestration, TrustRAG aims to provide a reliable, transparent, scalable, and publication-oriented intelligent document analysis framework for academic, industrial, and enterprise knowledge management.

Keywords: Retrieval-Augmented Generation (RAG), Adaptive Retrieval, Multi-Agent Systems, Large Language Models (LLMs), Explainable Artificial Intelligence (XAI), Evidence Verification, Intelligent Document Analysis, LangGraph, Qdrant, BM25, Dense Retrieval, Hybrid Retrieval, Information Retrieval, Generative AI, Knowledge Management, Confidence Estimation.

Domain: Artificial Intelligence (AI)

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